The normal distribution

1: Characteristics of the Normal Distribution

What shape is it? What parameters does it have?

2: Finding probabilities on a standard normal curve.

"Given that IQ is distributed as $X \sim N\left(100, 15^2\right)$, determine the probability that a randomly chosen person has an IQ above 130."

3: Finding unknown means/standard deviations.

In Wales, 30% of people have a height above 1.6m. Given the mean height is 1.4m and heights are normally distributed, determine the standard deviation of heights.

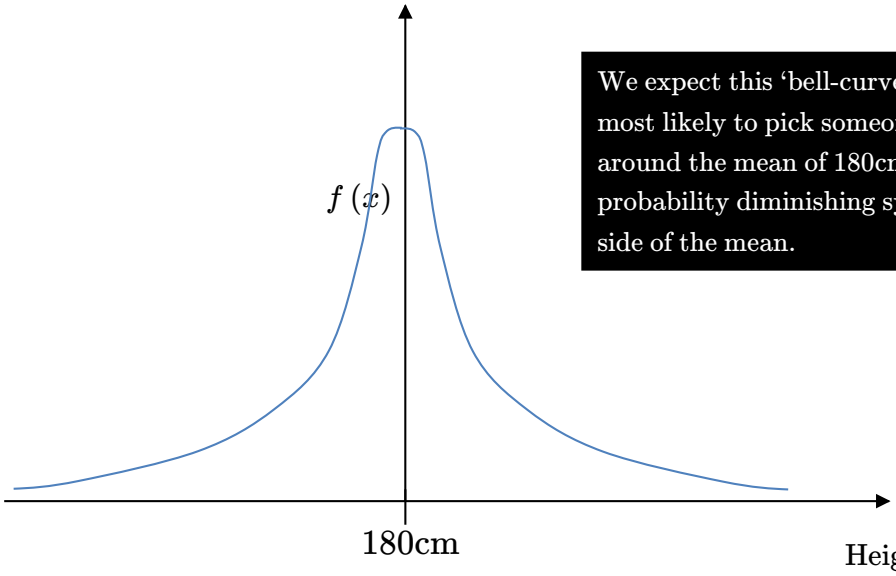
4: Binomial $\rightarrow$ Normal Approximations

How would I approximate $X \sim B(10, 0.4)$ using a Normal distribution? Under what conditions can we make such an approximation?

5: Hypothesis Testing

What does it look like? 3

The following shows what the probability distribution might look like for a random variable X, if X is the height of a randomly chosen person.

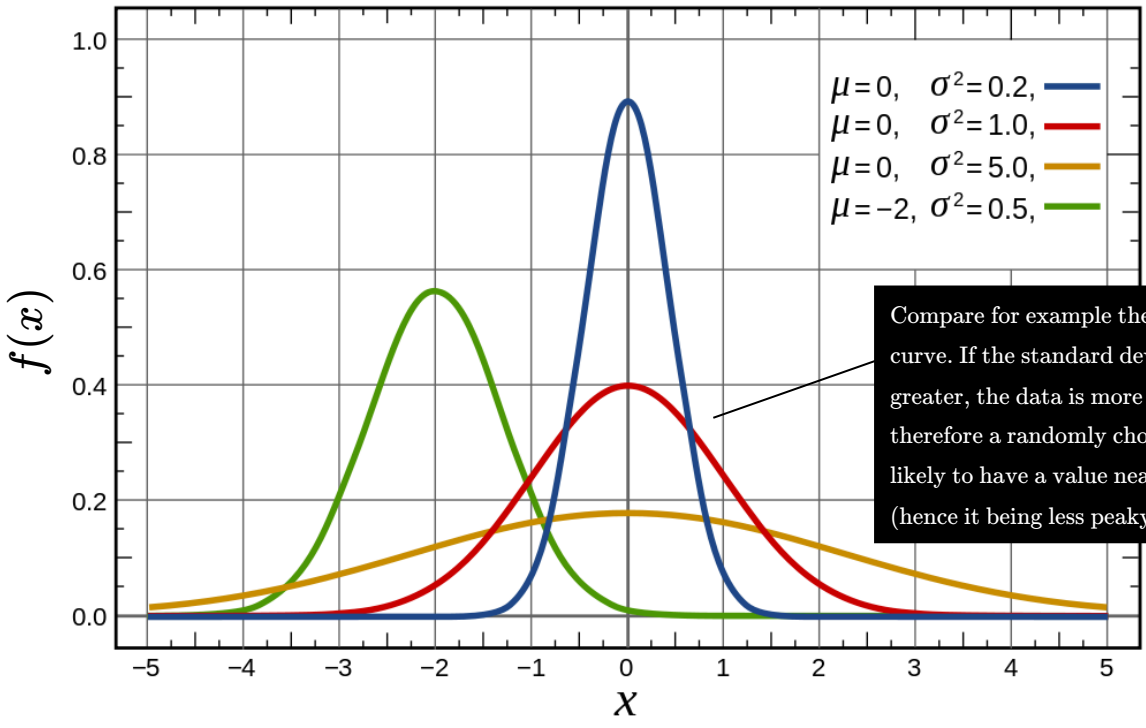


We expect this 'bell-curve' shape, where we're most likely to pick someone with a height around the mean of 180cm, with the probability diminishing symmetrically either side of the mean.

A variable with this kind of distribution is said to have a **normal distribution**.

For normal distributions we tend to draw the y axis at the mean for symmetry.

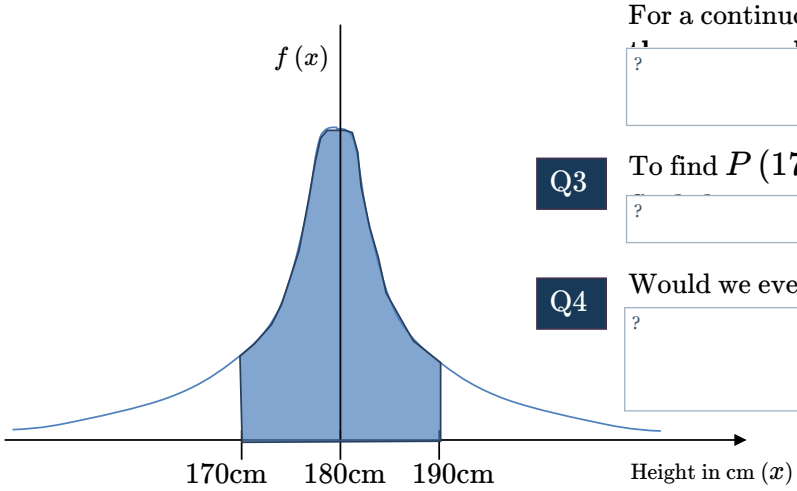
What does it look like? 4



Compare for example the blue and red curve. If the standard deviation is greater, the data is more spread out, and therefore a randomly chosen thing is less likely to have a value near the mean (hence it being less peaky).

We can set the mean μ and the standard deviation σ of the Normal Distribution. If a random variable X is normally distributed, then we write

$$\mathbf{X \sim N\left(\mu, \sigma^2\right)}$$



Q1 For a Normal Distribution to be used, the variable has to be:

?

Q2 With a discrete variable, all the probabilities had to add up to 1.
For a continuous variable, similarly:

?

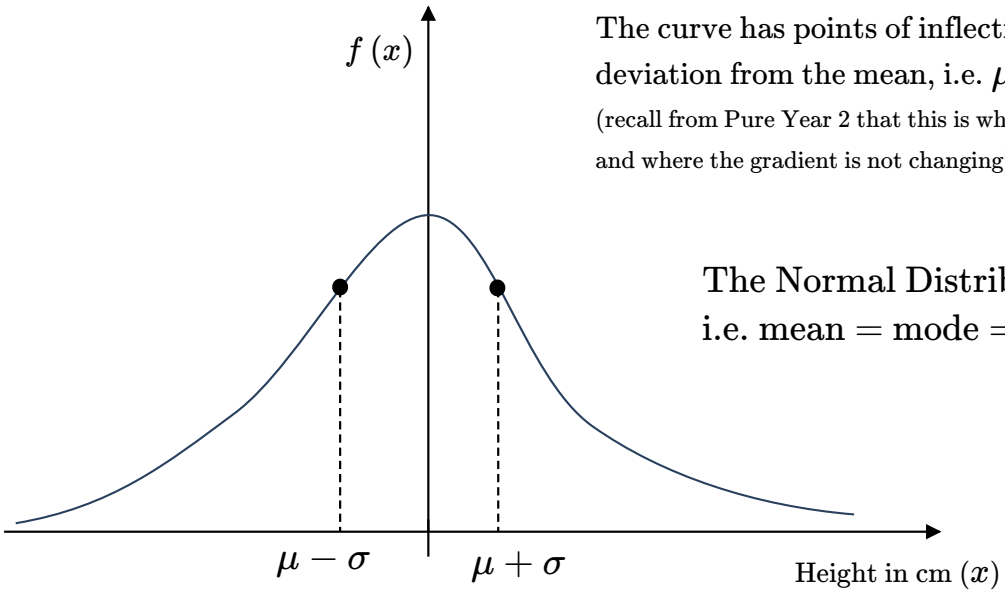
Q3 To find $P(170 < X < 190)$, we could:

?

Q4 Would we ever want to find $P(X = 200)$ say?

?

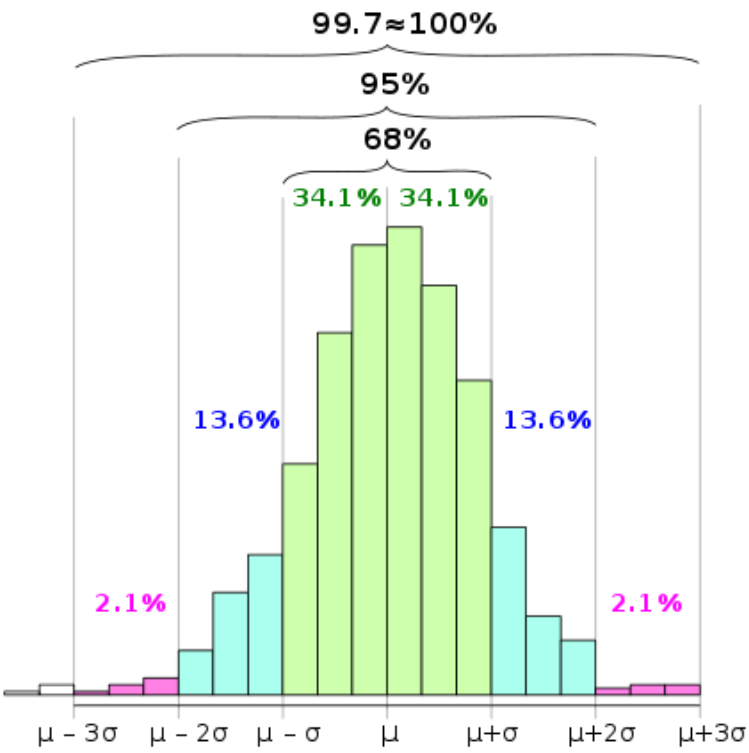
Side Notes: You might therefore wonder what the y-axis actually is. It is **probability density**, i.e. "the probability per unit cm". This is analogous to frequency density with histograms, where the y-value is frequency density area under the graph gives frequency. We use $f(x)$ rather than $p(x)$, to indicate probability density.



The curve has points of inflection one standard deviation from the mean, i.e. $\mu \pm \sigma$

(recall from Pure Year 2 that this is where the curve changes concavity, and where the gradient is not changing)

The Normal Distribution is symmetrical, i.e. mean = mode = median



The histogram above is for a quantity which is approximately normally distributed.

The 68-95-99.7 rule is a shorthand used to remember the percentage of data that is within 1, 2 and 3 standard deviations from the mean respectively.

You need to memorise this!

$\approx 68\%$ of data is within one standard deviation of the mean.
$\approx 95\%$ of data is within two standard deviations of the mean.
$\approx 99.7\%$ of data is within three standard deviations of the mean.

For practical purposes we consider all data to lie within $\mu \pm 5\sigma$

[Textbook] The diameters of a rivet produced by a particular machine, X mm, is modelled as $X \sim N\left(8, 0.2^2\right)$. Find:
a) $P\left(X > 8\right)$
b) $P\left(7.8 < X < 8.2\right)$

Tip: Draw a diagram!

a?

b?

a

IQ ("Intelligence Quotient") for a given population is, by definition, distributed using $X \sim N\left(100, 15^2\right)$. Find:

a) $P\left(70 < X < 130\right)$

b) $P\left(X > 115\right)$

Tip: Draw a diagram!

a?

b?

Exercise 3A – Statistics Y2 pages 40-41 10

6 The weights of a group of dormice, D grams, are modelled as $D \sim \mathrm{N}(\mu, 25)$. If 97.5% of dormice weigh less than 70 grams, find μ.

?

7 The masses of the pigs, M kg, on a farm are modelled as $M \sim \mathrm{N}(\mu, \sigma^2)$. If 84% of the pigs weigh more than 52 kg and 97.5% of the pigs weigh more than 47.5 kg, find μ and σ^2.

?

8 The percentage scores of a group of students in a test, S, are modelled as a normal distribution with mean 45 and standard deviation 15. Find:

a $P(S > 45)$ **b** $P(30 < S < 60)$ **c** $P(15 < S < 75)$

Alexia states that since it is impossible to score above 100%, this is not a suitable model.

d State, with a reason, whether Alexia is correct.

a?

b?

c?

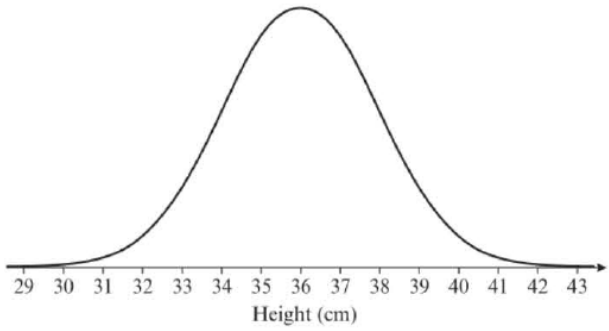
d?

9 The diagram shows the distribution of heights, in cm, of barn owls in the UK.

An ornithologist notices that the distribution is approximately normal.

a State the value of the mean height.

b Estimate the standard deviation of the heights.



a?

b?

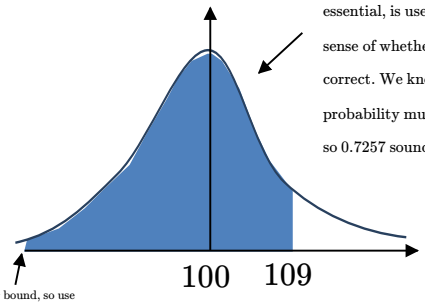
https://education.casio.co.uk/resources/videos/normal-distribution-probabilities-and-values

Just like a cumulative frequency graph gives the running total of the frequency up to a given value, a **cumulative distribution** gives the **running total of the probability** up to a given value.

IQ is distributed using $X \sim N\left(100, 15^2\right)$. Find
(a) $P\left(X < 109\right)$
(b) $P\left(X \geq 93\right)$
(c) $P\left(110 < X < 120\right)$
(d) $P\left(X < 80 or X > 106\right)$

a

1. Press MODE. Stats
2. Choose DISTRIBUTION
3. Choose Normal CD (i.e. "Cumulative Distribution")
4. Since the lower value is effectively $-\infty$, use any value at least 5σ below the mean (-100000 will do!). Press = after each value.
5. Put the upper value as 109.
6. Set $\sigma = 15$ and $\mu = 100$
7. You should obtain $P\left(X < 109\right) = 0.7257 (4dp)$



A diagram, while not essential, is useful to get a sense of whether our answer is correct. We know the probability must be above 0.5, so 0.7257 sounds about right.

No lower bound, so use -100000

IQ is distributed using $X \sim N\left(100, 15^2\right)$. Find
(a) $P\left(X < 109\right)$
(b) $P\left(X \geq 93\right)$
(c) $P\left(110 < X < 120\right)$
(d) $P\left(X < 80 or X > 106\right)$

b ?

Note: $\geq$ vs $>$ makes no difference as the distribution is continuous.

c ?

d ?

Test Your Understanding

The criteria for joining Mensa is an IQ of at least 131.

Assuming that IQ has the distribution $X \sim N\left(100, 15^2\right)$ for a population, determine:

a) What percentage of people are eligible to join Mensa.

b) If 30 adults are randomly chosen, the probability that at least 3 of them will be eligible to join. (Hint: Binomial distribution?)

a ?

b ?

Exercise 3B – Statistics Y2 pages 40-41

8 The volume of soap dispensed by a soap-dispenser on each press, X ml, is modelled as $X \sim \mathrm{N}(6, 0.8^2)$.

a Find: **i** $\mathrm{P}(X > 7)$ **ii** $\mathrm{P}(X < 5)$

The soap dispenser is pressed three times.

b Find the probability that on all three presses, less than 5 ml of soap is dispensed.

a?

b?

9 The amount of mineral water, W ml, in a bottle produced by a certain manufacturer is modelled as $W \sim \mathrm{N}(500, 14^2)$.

a Find: **i** $\mathrm{P}(W > 505)$ **ii** $\mathrm{P}(W < 490)$

A sample of 4 bottles is taken.

b Find the probability that all of the bottles contain more than 490 ml.

a?

b?

11 The diameters of bolts, D mm, made by a particular machine are modelled as $D \sim N(13, 0.1^2)$.

a Find the probability that a bolt, chosen at random, has a diameter less than 12.8 mm. **(1 mark)**

Bolts are considered to be 'perfect' if the diameter lies between 12.9 mm and 13.1 mm.
A random sample of 40 bolts is taken.

b Find the probability that more than 25 of the bolts are 'perfect'. **(4 marks)**

a?

b?

12 The masses, X grams, of a large population of squirrels are modelled as a normal distribution with $X \sim N(480, 40^2)$.

a Find the probability that a squirrel chosen at random has a mass greater than 490 g. **(1 mark)**

A naturalist takes a random sample of 30 squirrels from the population.

b Find the probability that at least 15 of the squirrels have a mass between 470 g and 490 g. **(4 marks)**

a?

b?

We now know how to use a calculator to value of the variable to obtain a probability. But we might want to do the reverse: given a probability of being in a region, how do we find the value of the boundary?

[Textbook] $X \sim N\left(20, 3^2\right)$. Find, correct to two decimal places, the values of a such that:

a. $P(X < a) = 0.75$

b. $P(X > a) = 0.4$

c. $P(16 < X < a) = 0.3$

?

?

?

If the IQ of a population is distributed using $X \sim N\left(100, 15^2\right)$.

a. Determine the IQ corresponding to the top 30% of the population.

b. Determine the interquartile range of IQs.

?

?

Test Your Understanding 21

$X \sim N\left(80, 7^2\right)$. Using your calculator,
a. determine the a such that $P(X > a) = 0.65$
b. determine the b such that $P(75 < X < b) = 0.4$
c. determine the c such that $P(c < X < 76) = 0.2$
d. determine the interquartile range of X.

a?

c?

b?

d?

5

Exercise 3C – Statistics Y2 pages 46-47 22

6 The masses, M kg, of a population of badgers are modelled as $M \sim \mathrm{N}(4.5, 0.6^2)$.
For this population, find:

a the lower quartile

b the 80th percentile

c Explain without calculation why $Q_2 = 4.5$ kg.

a?

b?

c?

7 The percentage scores, X, of a group of learner drivers in a theory test is modelled as a normal distribution with $X \sim N(72, 6^2)$.

a Find the value of a such that $P(X < a) = 0.6$. **(1 mark)**

b Find the interquartile range of the scores. **(2 marks)**

a?

b?

8 The masses, Y grams, of a brand of chocolate bar are modelled as $Y \sim N(60, 2^2)$.

a Find the value of y such that $P(Y > y) = 0.2$. **(1 mark)**

b Find the 10% to 90% interpercentile range of masses. **(2 marks)**

c Tom says that the median is equal to the mean. State, with a reason, whether Tom is correct. **(1 mark)**

a?

b?

c?

9 The distribution of heights, H cm, of a large group of men is modelled using $H \sim N(170, 10^2)$. A frock coat is a coat that goes from the neck of a person to near the floor. A clothing manufacturer uses the information to make three different lengths of frock coats. The table below shows the proportion of each size they will make.

Short	Regular	Long
30%	50%	20%

a The company wants to advertise a range of heights for which the regular frock coat is suitable. Use the model to suggest suitable heights for the advertisement. **(4 marks)**

b State one assumption you have made in deciding these values. **(1 mark)**

a?

b?

Z is the number of standard deviations above the mean.

Using the above definition, and considering the a variable (IQ) distributed as $X \sim N\left(100, 15^2\right)$,calculate (in your head!):

IQ	Z
100	?
130	?
85	?
165	?
62.5	?

What calculation are you actually doing in your head?

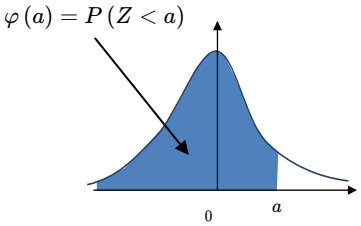
For an IQ of 130: $Z = \frac{130-100}{15} = 2$ as expected.

! Z represents the coding:
$Z = \frac{X-\mu}{\sigma}$
and $Z \sim N\left(0, 1^2\right)$. Z is known as a **standard** normal distribution.

The 0 and 1 of $Z \sim N\left(0, 1^2\right)$ are the result of the coding. If we've subtracted μ from each value the mean of the normal distribution is now 0. If we've divided all the values by σ the standard deviation is now $\frac{\sigma}{\sigma} = 1$

The point of coding in this context is that all different possible normal distributions become a single unified distribution where we no longer have to worry about the mean and standard deviation.
It means for example when we calculate $P\left(Z < 3\right)$, this will always give the same probability regardless of the original distribution.

It also means we can look up probabilities in a **z-table**:



$\Phi\left(a\right) = P\left(Z < a\right)$ is the cumulative distribution for the standard normal distribution. The values of $\Phi(a)$ can be found in a z-table.

This is a traditional z-table in the old A Level syllabus (but also found elsewhere). You no longer get given this and are expected to use your calculator.

THE NORMAL DISTRIBUTION FUNCTION

The function tabulated below is $\Phi(z)$, defined as $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{1}{2}t^2} \, dt$.

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.9772
0.01	0.5040	0.51	0.6950	1.01	0.8438	1.51	0.9345	2.02	0.9783
0.02	0.5080	0.52	0.6985	1.02	0.8461	1.52	0.9357	2.04	0.9793
0.03	0.5120	0.53	0.7019	1.03	0.8485	1.53	0.9370	2.06	0.9803
0.04	0.5160	0.54	0.7054	1.04	0.8508	1.54	0.9382	2.08	0.9812
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.10	0.9821
0.06	0.5239	0.56	0.7123	1.06	0.8554	1.56	0.9406	2.12	0.9830
0.07	0.5279	0.57	0.7157	1.07	0.8577	1.57	0.9418	2.14	0.9838
0.08	0.5319	0.58	0.7190	1.08	0.8599	1.58	0.9429	2.16	0.9846
0.09	0.5359	0.59	0.7224	1.09	0.8621	1.59	0.9441	2.18	0.9854
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.20	0.9861

This is from the new formula booklet. This is sometimes known as a 'reverse z-table', because you're looking up the z-value for a probability. Beware: p here it the probability of **exceeding** z rather than being up to z. Let's use it...

Percentage Points of The Normal Distribution

The values z in the table are those which a random variable $Z \sim N(0, 1)$ exceeds with probability p; that is, $P(Z > z) = 1 - \Phi(z) = p$.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Examples

[Textbook] The random variable $X \sim N\left(50, 4^2\right)$. Write in terms of $\Phi\left(z\right)$ for some value of z.

(a) $P\left(X < 53\right)$ (b) $P\left(X \geq 55\right)$

a ?

b ?

[Textbook] The systolic blood pressure of an adult population, S mmHg, is modelled as a normal distribution with mean 127 and standard deviation 16. A medical research wants to study adults with blood pressures higher than the 95th percentile. Find the minimum blood pressure for an adult included in her study.

?

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Further Examples

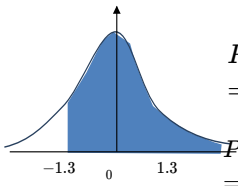
(a) Determine $P(Z > -1.3)$

(b) Determine $P\left(-2 < Z < 1\right)$

(c) Determine the a such that $P\left(Z > a\right) = 0.7$

(d) Determine the a such that $P\left(-a < Z < a\right) = 0.6$

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

a 

$P\left($?
$= 0$

b $P\left(-$?
$= P$
$= 0.$
$= 0.$ -2)

Tip: Either changing $<$ to/from $>$ or changing the sign ($+$ to/from -) has the effect of "$1-$". However, if you change both, the "$1-$"s cancel out!

$P\left(a < Z < b\right) = P\left(Z < b\right) - P\left(Z < a\right)$

c ?

d ?

Test Your Understanding

1 IQ is distributed with mean 100 and standard deviation 15. Using an appropriate table, determine the IQ corresponding to the
(a) top 10% of people.
(b) bottom 20% of people.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

a ?

b ?

2 If $X \sim N\left(100, 15^2\right)$, determine, in terms of Φ:
(a) $P(X > 115)$
(b) $P\left(77.5 < X < 112\right)$

a ?

b ?

3 Find the a such that:
(a) $P\left(-a < Z < a\right) = 0.2$
(b) $P\left(0 < Z < a\right) = 0.35$

a ?

b ?

Exercise 3D – Statistics Y2 pages 49

5 a Use the percentage points table to find a value of z such that $P(Z > z) = 0.025$. **(1 mark)**

b A fighter jet training programme takes only the top 2.5% of candidates on a test. Given that the scores can be modelled using a normal distribution with mean 80 and standard deviation 4, use your answer to part **a** to find the score necessary to get on the programme. **(2 marks)**

a?

b?

6 a Use the percentage points table to find a value of z such that $P(Z < z) = 0.15$. **(1 mark)**

b A hat manufacturer makes a special 'petite' hat which should fit 15% of its customers. Given that hat sizes can be modelled using a normal distribution with mean 57 cm and standard deviation 2 cm, use your answer to part **a** to find the size of a 'petite' hat. **(2 marks)**

a?

b?

7 a Use the percentage points table to find the values of z that correspond to the 10% to 90% interpercentile range. **(2 marks)**

A particular brand of light bulb has a life modelled as a normal distribution with mean 1175 hours and standard deviation 56 hours. The bulb life is considered 'standard' if its life falls into the 10% to 90% interpercentile range.

b Use your answer to part **a** to find the range of life to the nearest hour for a 'standard' bulb. **(2 marks)**

a?

b?

Finding the missing mean or standard deviation

In the last section, you may have thought, "what's the point of standardising to Z when I can just use the DISTRIBUTION mode on my calculator?"

HINT: both forward and reverse normal lookups on the calculator **required you to specify μ and σ.**

[Textbook] $X \sim N\left(\mu, 3^2\right)$. Given that $P\left(X > 20\right) = 0.2$, find the value of μ.

?

In the last section, you may have thought, "what's the point of standardising to Z when I can just use the DISTRIBUTION mode on my calculator?"

[Textbook] A machine makes metal sheets with width, X cm, modelled as a normal distribution such that $X \sim N\left(50, \sigma^2\right)$.

(a) Given that $P\left(X < 46\right) = 0.2119$, find the value of σ.

(b) Find the 90th percentile of the widths.

?

The method here is exactly the same as before:

1. Using a sketch, determine whether you're expecting a positive or negative z value.
2. Look up z value, using tables if you can (otherwise your calculator). Make negative if in bottom half.
3. Use $Z = \frac{X-\mu}{\sigma}$

When both are missing

If both μ and σ are missing, we end up with simultaneous equations which we must solve.

Edexcel S1 Jan 2011

The weight, Y grams, of soup put into a carton by machine B is normally distributed with mean μ grams and standard deviation σ grams.

(c) Given that $\mathrm{P}(Y < 160) = 0.99$ and $\mathrm{P}(Y > 152) = 0.90$, find the value of μ and the value of σ. **(6)**

?

Edexcel S1 May 2013 (R)

The time taken to fly from London to Berlin has a normal distribution with mean 100 minutes and standard deviation d minutes.

Given that 15% of the flights from London to Berlin take longer than 115 minutes,

(b) find the value of the standard deviation d.

(4)

?

Edexcel S1 Jan 2002

5. The duration of the pregnancy of a certain breed of cow is normally distributed with mean μ days and standard deviation σ days. Only 2.5% of all pregnancies are shorter than 235 days and 15% are longer than 286 days.

(a) Show that $\mu - 235 = 1.96\sigma$.

(2)

(b) Obtain a second equation in μ and σ.

(3)

(c) Find the value of μ and the value of σ.

(4)

(d) Find the values between which the middle 68.3% of pregnancies lie.

(2)

(a) ?

(b) ?

(c) ?

(d) ?

11 An automated pottery wheel is used to make bowls. The diameter of the bowls, D mm, is normally distributed with mean μ and standard deviation 5 mm. Given that 75% of bowls are greater than 200 mm in diameter, find:

a the value of μ **(2 marks)**

b P($204 < D < 206$) **(1 mark)**

Three bowls are chosen at random.

c Find the probability that all of the bowls are greater than 205 mm in diameter. **(3 marks)**

a?

b?

c?

12 A loom makes table cloths with an average thickness of 2.5 mm. The thickness, T mm, can be modelled using a normal distribution. Given that 65% of table cloths are less than 2.55 mm thick, find:

a the standard deviation of the thickness **(2 marks)**

b the proportion of table cloths with thickness between 2.4 mm and 2.6 mm. **(1 mark)**

A table cloth can be sold if the thickness is between 2.4 mm and 2.6 mm. A sample of 20 table cloths is taken.

c Find the probability that at least 15 table cloths can be sold. **(3 marks)**

a?

b?

c?

13 The masses of the penguins on an island are found to be normally distributed with mean μ, and standard deviation σ. Given that 10% of the penguins have a mass less than 18 kg and 5% of the penguins have a mass greater than 30 kg,

a sketch a diagram to represent this information **(2 marks)**

b find the value of μ and the value of σ. **(6 marks)**

10 penguins are chosen at random.

c Find the probability that at least 4 of them have a mass greater than 25 kg. **(4 marks)**

a?

b?

c?

14 The length of an adult Dachshund is found to be normally distributed with mean μ and standard deviation σ. Given that 20% of Dachshunds have a length less than 16 inches and 10% have a length greater than 18 inches, find:

a the value of μ and the value of σ **(6 marks)**

b the interquartile range. **(2 marks)**

a?

b?

A Normally distributed random variable $X \sim N(\mu, \sigma^2)$ has interquartile range q.

a Show that $\sigma = 0.742q$, where the coefficient q is correct to 3 s.f.

b Explain why it is not possible to write μ in terms of q only.

a?

b?

The binomial distribution

The number of successes in a fixed number of trials

X~B(*n*,*p*) *n* the number of trials
p the probability of success

Properties

?

?

?

?

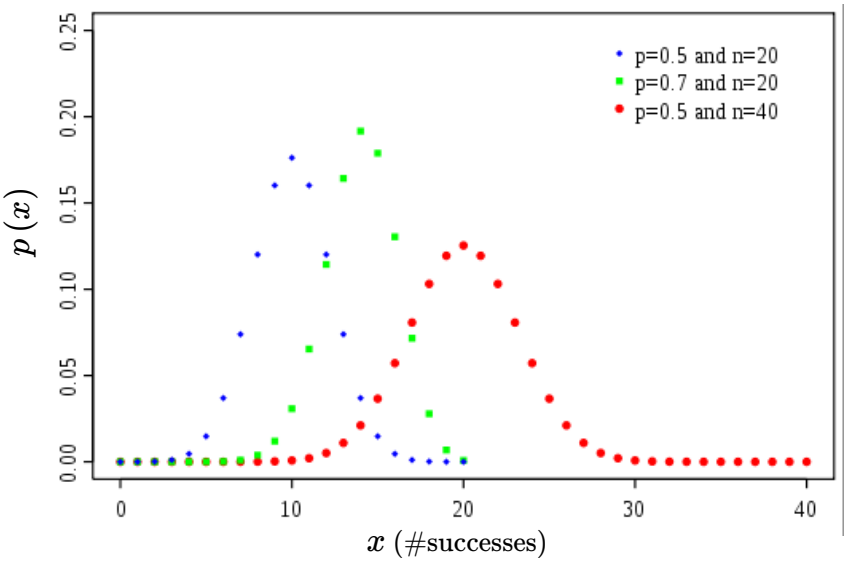
Calculation

If $X \sim B(n,p)$ then: $\mathbf{P}(\mathbf{X}=\mathbf{r}) = \binom{\mathbf{n}}{\mathbf{r}}$?

The binomial distribution - Examples

- The number of flawed garments in a sample of 50
- The number of people who voted in favour of a new initiative in a sample of 3000
  (assume the people are not related so the trials are independent)
- The number of students who are left handed in a class of 34

Approximating a Binomial Distribution



The graph shows the probability function for different Binomial Distributions. Which one resembles another distribution and what distribution does it resemble?

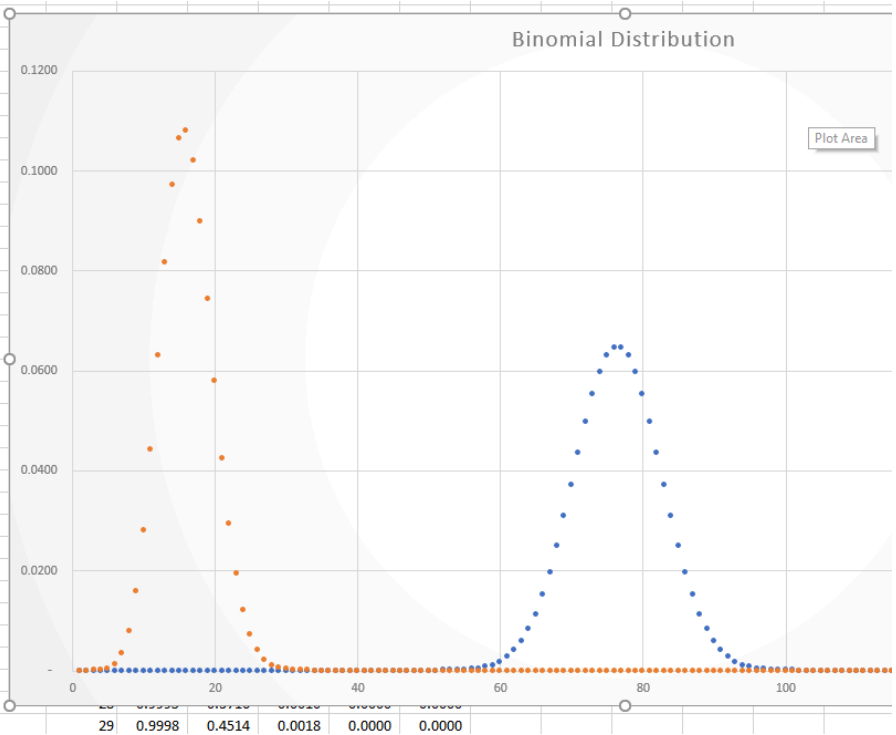
?

Modelling binomial distribution in excel

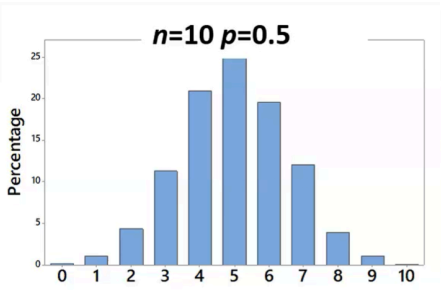
Background curiosity.. How to plot a binomial distribution in Excel

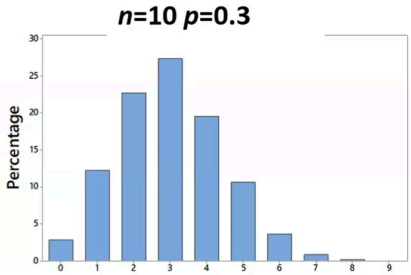
Microsoft Excel Worksheet

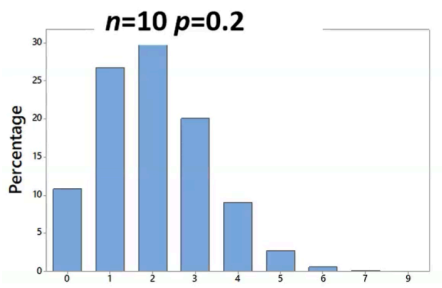
r	Normal 0.1	0.2	0.3	0.4	0.5
0	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0001	0.0000	0.0000	0.0000	0.0000
4	0.0004	0.0000	0.0000	0.0000	0.0000
5	0.0013	0.0000	0.0000	0.0000	0.0000
6	0.0035	0.0000	0.0000	0.0000	0.0000
7	0.0079	0.0000	0.0000	0.0000	0.0000
8	0.0159	0.0000	0.0000	0.0000	0.0000
9	0.0280	0.0000	0.0000	0.0000	0.0000
10	0.0443	0.0000	0.0000	0.0000	0.0000
11	0.0630	0.0000	0.0000	0.0000	0.0000
12	0.0817	0.0000	0.0000	0.0000	0.0000
13	0.0971	0.0001	0.0000	0.0000	0.0000
14	0.1063	0.0002	0.0000	0.0000	0.0000
15	0.1079	0.0004	0.0000	0.0000	0.0000
16	0.1019	0.0008	0.0000	0.0000	0.0000
17	0.0899	0.0016	0.0000	0.0000	0.0000
18	0.0744	0.0031	0.0000	0.0000	0.0000
19	0.0578	0.0054	0.0000	0.0000	0.0000
20	0.0424	0.0089	0.0000	0.0000	0.0000
21	0.0294	0.0138	0.0000	0.0000	0.0000
22	0.0193	0.0204	0.0000	0.0000	0.0000
23	0.0120	0.0286	0.0000	0.0000	0.0000
24	0.0071	0.0381	0.0000	0.0000	0.0000
25	0.0040	0.0484	0.0001	0.0000	0.0000
26	0.0022	0.0587	0.0001	0.0000	0.0000
27	0.0011	0.0679	0.0002	0.0000	0.0000
28	0.0005	0.0752	0.0005	0.0000	0.0000
29	0.0003	0.0797	0.0009	0.0000	0.0000
29	0.9998	0.4514	0.0018	0.0000	0.0000



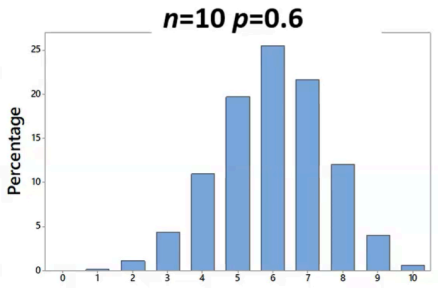
As p decreases the distribution becomes more positively skewed

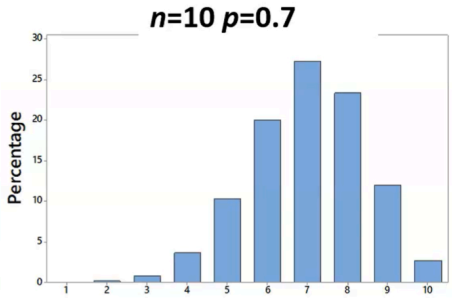


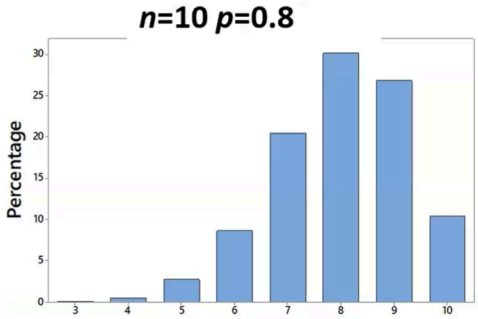




As p increases the distribution becomes more negatively skewed







If we're going to use a normal distribution to approximate a Binomial distribution, it makes sense that we set the mean and standard deviation of the normal distribution to match that of the original binomial distribution:

$\mu = np$?

$\sigma = \sqrt{np(}$?

If n is large and p close to 0.5, then the binomial distribution $X \sim B(n, p)$ can be approximated by the normal distribution $N(\mu, \sigma^2)$ where

$\mu = np$

$\sigma = \sqrt{np(1-p)}$

Quickfire Questions:

$X \sim B(10, 0.2) \rightarrow Y$?

$X \sim B(20, 0.5) \rightarrow Y$?

$X \sim B(6, 0.3) \rightarrow Y$? 5)

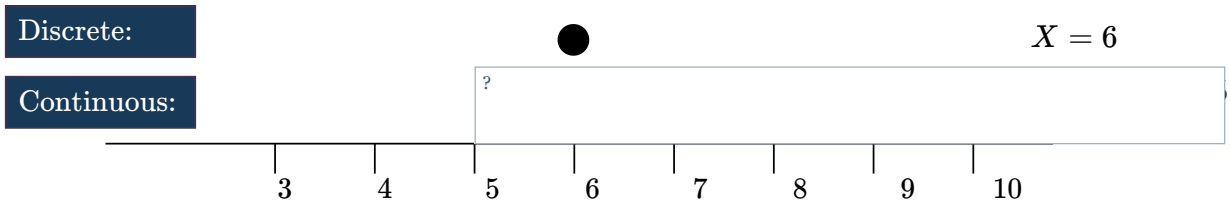
We tend to use the letter Y to represent the normal distribution approximation of the distribution X.

Why use a normal approximation?

- Tables for the binomial distribution only goes up to $n = 50$ and your calculator will reject large values of n.
- The formula for $P(X = x)$ makes use a factorials. Factorials of large numbers cannot be computed efficiently. Type in 65! for example; your calculator will hesitate! Now imagine how many factorials would be required if you wanted to find $P(X \leq 65)$.

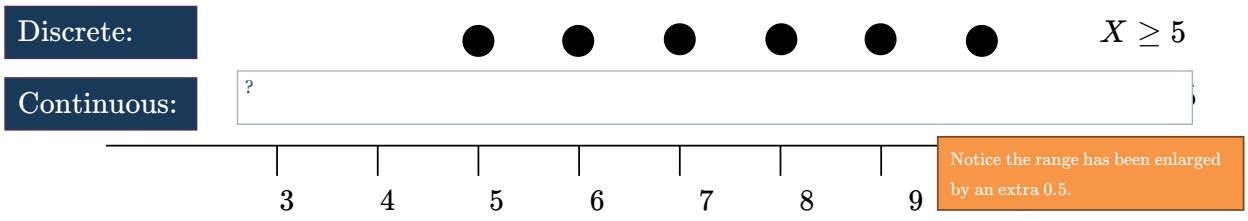
One problem is that the outcomes of a binomial distribution (i.e. number of successes) are **discrete** whereas the Normal distribution is **continuous**.
We apply something called a **continuity correction** to approximate a discrete distribution using a continuous one.

The random variable X represents the time to finish a race in hours. We're interested in knowing the probability Alice took 6 hours to the nearest hour.
How would you represent this time on a number line given hours is discrete? And what about if hours was now considered to be continuous (as Y)?

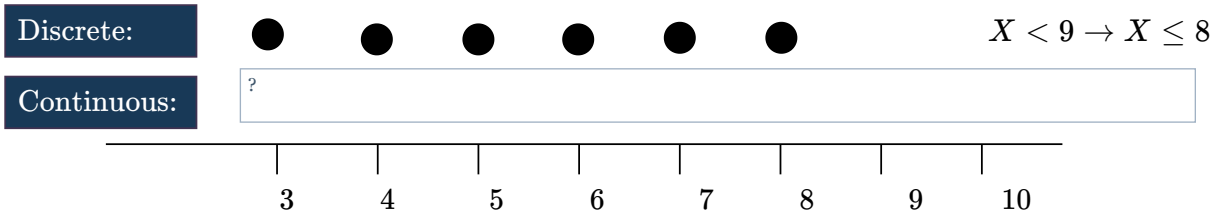


We can't just find $\mathbf{P}(\mathbf{Y} = \mathbf{6})$ when $\mathbf{Y}$ is continuous, because the probability is effectively 0. But $\mathbf{P}(\mathbf{5.5} < \mathbf{Y} < \mathbf{6.5})$ would seem a sensible interval to use because any time between 5.5 and 6.5 would have rounded to 6 hours were it discrete.

If X is a discrete variable, and Y is its continuous equivalent, how would you represent $P(X \geq 5)$ for Y?



How would represent $P(X < 9)$ for Y?



A continuity correction is approximating a discrete range using a continuous one.
1. If $>$ or $<$, convert to $\geq$, $\leq$ first.
2. Enlarge the range by 0.5.

Examples

Discrete	→	Continuous
$P(X \leq 7)$		?
$P(X < 10)$		?
$P(X > 9)$		?
$P(1 \leq X \leq 10)$		?
$P(3 < X < 6)$		?
$P(3 \leq X < 6)$		?
$P(3 < X \leq 6)$		?
$P(X = 3)$		?

A continuity correction is approximating a discrete range using a continuous one.
1. If $>$ or $<$, convert to $\geq$, $\leq$ first.
2. Enlarge the range (at each end) by 0.5.

Comparison on your calculators

400 young children guess the answer true or false to 80 quantum physics questions. What proportion of children will get less than 38 questions correct?

1. Find Normal Distribution using n and p

2. Adjust for continuity correction

400 young children guess the answer true or false to 80 quantum physics questions. What proportion of children will get less than 38 questions correct?

Normal Distribution	Binomial Distribution
Write distribution	Write distribution
Calculate on fx-CG50	Calculate on fx-CG50
Display result	Display result

[Textbook - Edited] For a particular type of flower bulbs, 55% will produce yellow flowers. A random sample of 80 bulbs is planted.
(a) Calculate the actual probability that there are exactly 50 flowers.
(b) Use a normal approximation to find a estimate that there are exactly 50 flowers.
(c) Hence determine the percentage error of the normal approximation for 50 flowers.

a ?

b ?

c ?

Edexcel S2 Jan 2004 Q3

The discrete random variable X is distributed B(n, p).

(*a*) Write down the value of p that will give the most accurate estimate when approximating the binomial distribution by a normal distribution. **(1)**

(*b*) Give a reason to support your value. **(1)**

(*c*) Given that $n = 200$ and $p = 0.48$, find $P(90 \leq X < 105)$. **(7)**

a?

b?

c?

2 The random variable $X \sim B(150, 0.45)$. Use a suitable approximation to estimate:

a $P(X \leqslant 60)$? **b** $P(X > 75)$? **c** $P(65 \leqslant X \leqslant 80)$?

3 The random variable $X \sim B(200, 0.53)$. Use a suitable approximation to estimate:

a $P(X < 90)$? **b** $P(100 \leqslant X < 110)$? **c** $P(X = 105)$?

4 The random variable $X \sim B(100, 0.6)$. Use a suitable approximation to estimate:

a $P(X > 58)$? **b** $P(60 < X \leqslant 72)$? **c** $P(X = 70)$?

5 A fair coin is tossed 70 times. Use a suitable approximation to estimate the probability of obtaining more than 45 heads. ?

6 The probability of a roulette ball landing on red when the wheel is spun is $\frac{50}{101}$.
On one day in a casino, the wheel is spun 1200 times.
Estimate the probability that the ball lands on red in at least half of these spins.

?

7 **a** Write down two conditions under which the normal distribution may be used as an approximation to the binomial distribution. **(2 marks)**

A company sells orchids of which 45% produce pink flowers.

A random sample of 20 orchids is taken and X produce pink flowers.

b Find $P(X = 10)$. **(1 mark)**

A second random sample of 240 orchids is taken.

c Using a suitable approximation, find the probability that fewer than 110 orchids produce pink flowers. **(3 marks)**

d The probability that at least q orchids produce pink flowers is 0.2. Find q. **(3 marks)**

a?

b?

c?

d?

8 A drill bit manufacturer claims that 52% of its bits last longer than 40 hours.
A random sample of 30 bits is taken and X last longer than 40 hours.

a Find $P(X < 17)$. **(1 mark)**

A second random sample of 600 drill bits is taken.

b Using a suitable approximation, find the probability that between 300 and 350 bits last longer than 40 hours. **(3 marks)**

a?

b?

9 A particular breakfast cereal has prizes in 56% of the boxes. A random sample of 100 boxes is taken.

a Find the exact value of the probability that exactly 55 boxes contain a prize. **(1 mark)**

b Find the percentage error when using a normal approximation to calculate the probability that exactly 55 boxes contain prizes. **(4 marks)**

a?

b?

Suppose you used a random number generator to choose three numbers at random from the integers 1 – 100 inclusive.

e.g. 55, 74, 5

What is the mean of this sample? $\overline{x} =$?

Smallest possible sample mean is ?

Largest possible sample mean is ?

Which sample means are more likely?

If we worked out the probability for each possible sample mean, what would the distribution look like?

Roll of a fair die: sample size 1

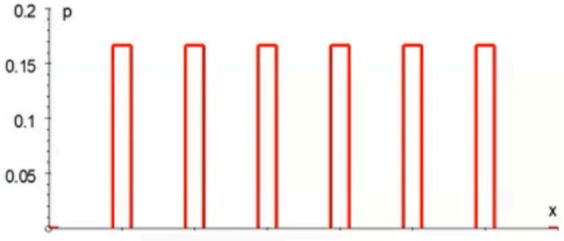
Roll an ordinary fair die once.

The population is {1, 2, 3, 4, 5, 6}

The possible samples of size 1 are ?

The probability distribution of the sample means is

$\overline{x}$	?
$p(\overline{X} = \overline{x})$	



It can be shown that

$E(\overline{X}) =$?

$V(\overline{X}) =$?

Note: The mean and variance of the sample means are equal to those for the population {1, 2, 3, 4, 5, 6}.

Roll an ordinary fair die twice.

How many possible samples are there of size 2?

What are the possible sample means?

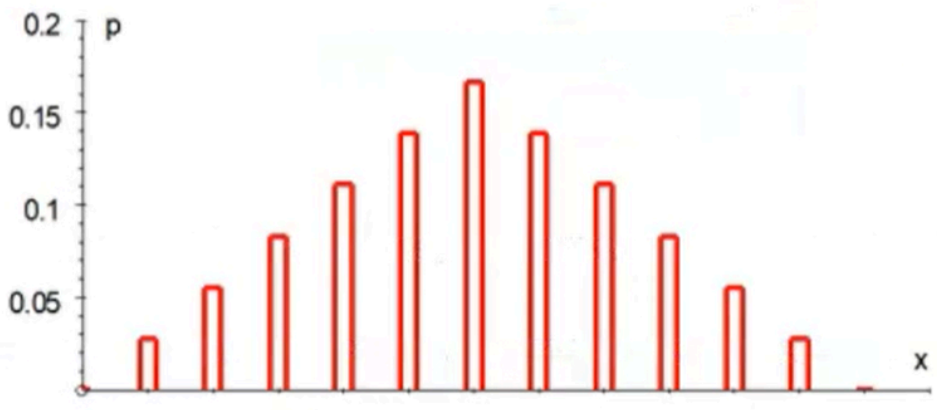
	1	2	3	4	5	6
1	?					
2						
3						
4						
5						
6						

The probability distribution of the sample means is

$\bar{y}$	?
$P(\bar{Y} = \bar{y})$	

The probability distribution of the sample means is

$\bar{y}$	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
$P(\bar{Y} = \bar{y})$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

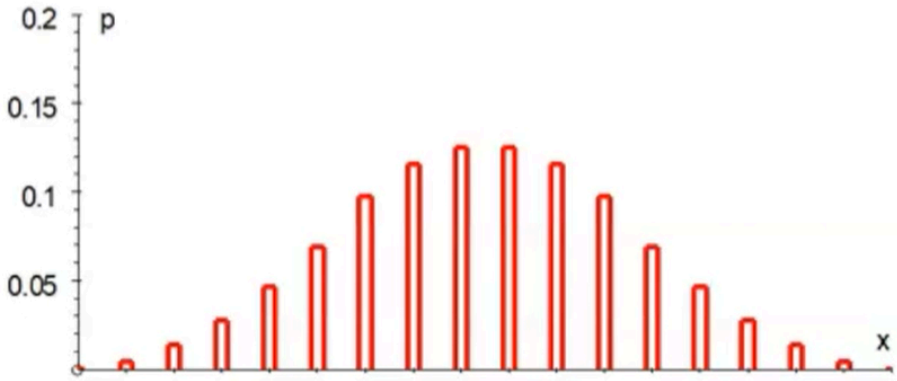


$E(\bar{Y}) =$?

$V(\bar{Y}) =$?

The probability distribution of the sample means is

$\overline{z}$	?
$P(\overline{Z} = \overline{z})$	



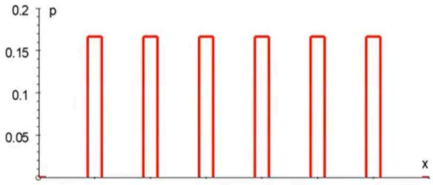
It can be shown that

$E(\overline{Z}) =$?

$V(\overline{Z}) =$?

What did you notice as we increased the sample size n?

$n = 1$



?

$n = 2$

?

$n = 3$

?

- There is a pattern in the mean and standard deviation of the sample means which depends on n

Sample size n	Mean	Variance	Standard deviation
1	?	?	?
2			
3			

Patterns relating to the sample sizes

What did you notice as we increased the sample size n?

- There is a pattern in the mean and standard deviation of the sample means which depends on n

Sample size n	Mean	Variance	Standard deviation
1	3.5	$\frac{35}{12}$	$\sqrt{\frac{35}{12}}$
2	3.5	$\frac{35}{24}$	$\sqrt{\frac{35}{24}}$
3	3.5	$\frac{35}{36}$	$\sqrt{\frac{35}{36}}$

← **Sample size 1**
Mean $= \mu$, SD $= \sigma$

← **Sample size 2**
Mean $= \mu$, SD $= \sigma/\sqrt{2}$

← **Sample size 3**
Mean $= \mu$, SD $= \sigma/\sqrt{3}$

Summary

Given a population with mean μ and standard deviation σ

- The distribution of the sample means also has mean μ
- The standard deviation of the sample means (sometimes called the *standard error of the mean*) is smaller than the population standard deviation and it decreases as n increases. It has the value $\frac{\sigma}{\sqrt{n}}$

For a random sample of size n from a normally distributed population X, the sample mean $\overline{X}$ is normally distributed with $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

Let's assume the underlying distribution of the population is normal $N\left(\mu, \sigma^2\right)$

When a sample is taken of size n the sample mean $\left(\overline{X}\right)$ is normally distributed with $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

Standardising the distribution of the sample means

Normal distribution	Distribution of the sample means
$X \sim N(\mu, \sigma^2)$	?
$z = \frac{x - \mu}{\sigma}$	?

Calculator skills (Casio fx-CG50): Normal hypothesis testing

4 steps for setting up a Hypothesis Test for the population mean:

1. The null hypothesis is $\mu = m$ where m is a suggested value of the population mean

2. The alternative hypothesis is one of:
   - $\mu > m$ (one sided test)
   - $\mu < m$ (one sided test)
   - $\mu \neq m$ (two sided test)

3. Determine the significance level of the test

4. Consider the evidence from the sample

Test results are normally distributed with a mean of 65 and a standard deviation of 10. After the introduction of a dynamic new teacher the results for a group of 8 students had a mean of 72. Is there evidence that the results have significantly improved at a 5% level of significance?

$X \sim$?

H_0: ?

H_1: ?

where μ is the population mean score

$\overline{X} \sim$?

The distribution of the sample mean is $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

We need to find $p(\overline{X} > 72) = p\left(Z > \text{?}\right)$

$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$p(Z >$? $) =$?

Test results are normally distributed with a mean of 65 and a standard deviation of 10. After the introduction of a dynamic new teacher the results for a group of 8 students had a mean of 72. Is there evidence that the results have significantly improved at a 5% level of significance?

In summary:

$H_0: \mu = 65$

$H_1: \mu > 65$

where μ is the population mean score

$\overline{X} \sim N\left(65, \frac{10^2}{8}\right)$

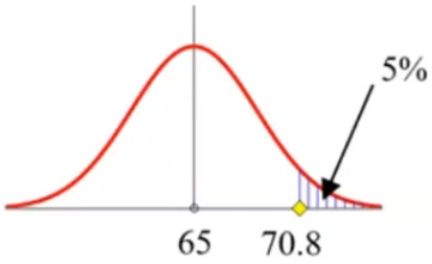
$p(\overline{X} > 72) = 0.0239$

$0.0239 < 0.05$ so there is evidence to reject the null hypothesis and conclude that the mean score has increased.

Test results are normally distributed with a mean of 65 and a standard deviation of 10. After the introduction of a dynamic new teacher the results for a group of 8 students had a mean of 72. Is there evidence that the results have significantly improved at a 5% level of significance?

$$\overline{X} \sim N\left(65, \frac{10^2}{8}\right)$$

Using your calculator you find that the critical region is $\bar{x} > 70.8$



The evidence given is a sample mean $\bar{x} = 72$. This lies in the critical region so we conclude there is evidence to reject H_0 and that the mean score has increased.

The supplier of LITE light bulbs claims that the mean life of a LITE light bulb is 130 hours. Responding to customer complaints that the light bulbs did not last as long as expected, a trading standards organisation tested 400 bulbs and found the mean to be 128.5 hours and the sample standard deviation was 13 hours.
Is there evidence at a 2% level that the mean is lower than 130 hours?

Let X be the distribution of times of LITE light bulbs

Then $\overline{X} \sim N\left(\boxed{?}, \boxed{?}\right)$

$H_0: \mu$?

$H_1: \mu$

$p(\overline{X} < 128.5) =$?

Conclusion: ?

Example 2: Left Hand Tail using a critical region

The supplier of LITE light bulbs claims that the mean life of a LITE light bulb is 130 hours. Responding to customer complaints that the light bulbs did not last as long as expected, a trading standards organisation tested 400 bulbs and found the mean to be 128.5 hours and the sample standard deviation was 13 hours.
Is there evidence at a 2% level that the mean is lower than 130 hours?

1. The setup of the test is identical:

Let X be the distribution of times of LITE light bulbs

Then $\bar{X} \sim N\left(130, \frac{13^2}{400}\right)$ $H_0: \mu = 130$ $H_1: \mu < 130$

2. Using your calculator

The inverse Normal for 0.02 is [?] so the critical region is $\bar{X} <$ [?]

Conclusion: [?]

Left Hand Tail: Method 2 using a critical region

The supplier of LITE light bulbs claims that the mean life of a LITE light bulb is 130 hours. Responding to customer complaints that the light bulbs did not last as long as expected, a trading standards organisation tested 400 bulbs and found the mean to be 128.5 hours and the sample standard deviation was 13 hours.
Is there evidence at a 2% level that the mean is lower than 130 hours?

1. The setup of the test is identical:

Let X be the distribution of times of LITE light bulbs

Then $\bar{X} \sim N\left(130, \frac{13^2}{400}\right)$ $H_0: \mu = 130$ $H_1: \mu < 130$

2. Using your calculator

The inverse Normal for 0.02 is [?] so the critical region is $\bar{X} <$ [?]

Conclusion: [?]

Two tailed tests

In previous examples we have looked at examples of one-tailed tests where the alternative hypothesis was of the form;

$$\mu < k \text{ or } \mu > k$$

Sometimes the alternative hypothesis needs to be of the form

$$\mu \neq k$$

Here we are not stating in which direction the mean may lie, just that it might not be the stated value.

There are two parts to the critical region (both tail of the distribution are considered)

- At the 5% significant level the critical region will have 2.5% in each tail
- When considering the p values compare the value with half of the significance level as you are just comparing with one of the tails

Two tailed test example

The lengths of the leaves of a certain species of rare plant are Normally distributed with mean 8.6 cm and standard deviation 1.2 cm. A botanist finds a clump of plants and wants to find out whether they are of the rare species. She collects and measures 50 leaves and finds that the total of their lengths is 442 cm. Carry out a test at the 5% level. What should the botanist conclude?

X is the distribution of the lengths of leaves

$$\bar{X} \sim N\left(8.6, \frac{1.2^2}{50}\right)$$

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$H_0: \mu = 8.6 \qquad H_1: \mu \neq 8.6$

We need a tests statistic from the data collected. If the total length of the 50 leaves is 442cm the sample mean is $\bar{x} = \frac{442}{50} = 8.84$ cm

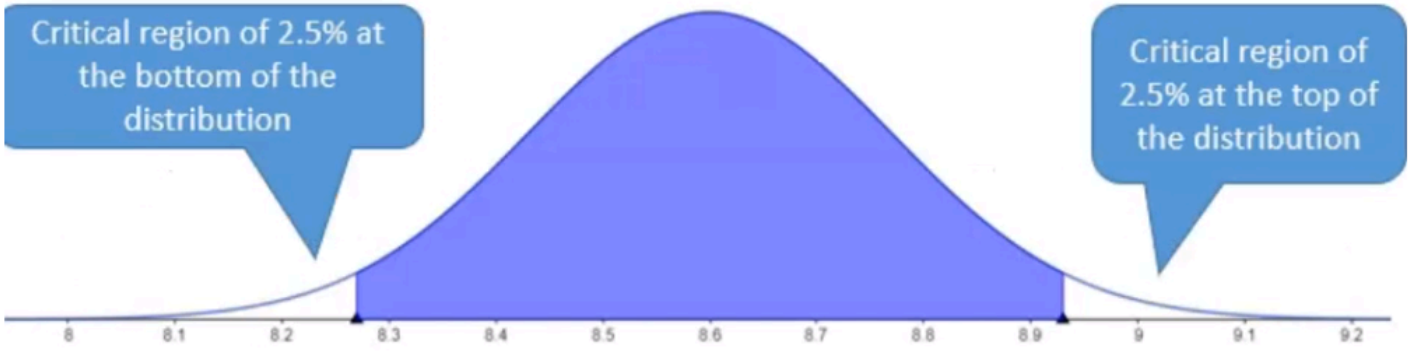
$$p(\bar{X} > 8.84) = 1 - p(\bar{X} < 8.84)$$
$$= 1 - 0.92135.. = 0.0787$$

As 0.0787 > 0.025 there is no evidence to reject the null hypothesis. We cannot conclude that the plants are not of the rare species.

Note: Using the inverse normal of 0.975 gives a value of 8.93 so the critical region is $\overline{X} > 8.93$
So 8.84<8.93 so there is no evidence to reject the null hypothesis

The lengths of the leaves of a certain species of rare plant are Normally distributed with mean 8.6 cm and standard deviation 1.2 cm. A botanist finds a clump of plants and wants to find out whether they are of the rare species. She collects and measures 50 leaves and finds that the total of their lengths is 442 cm. Carry out a test at the 5% level. What should the botanist conclude?

$$\bar{X} \sim N\left(8.6, \frac{1.2^2}{50}\right) \qquad H_0: \mu = 8.6 \qquad H_1: \mu \neq 8.6$$



1. For each of the following, a random sample of size n is taken from a Normal population with known standard deviation. The sample mean $\bar{x}$ is calculated.
In each case find the p-value and carry out the hypothesis tests, given H_0 and H_1 at the significance level indicated.

	σ	n	$\bar{x}$	H_0	H_1	Sig. level
(i)	14	20	247	$\mu = 240$	$\mu \neq 240$	2%
(ii)	15	10	172.9	$\mu = 165$	$\mu > 165$	5%
(iii)	24	15	226	$\mu = 240$	$\mu < 240$	1%

(i) ?

(ii) ?

(iii) ?

Exercise 2

The masses of adult students are known to be Normally distributed with mean 67.4 kg and standard deviation 3.8 kg. A sample of size 24 is taken and the mean found to be 65.8 kg. Assuming that the standard deviation is unchanged, test, at the 1% significance level, whether the mean mass of adult students has decreased, giving the critical region for the test.

?

Exercise 3

Ball bearings produced by a machine should have diameters of 15 mm. A random sample of the diameters of 80 ball bearings gave a sample mean of 14.64 mm. The standard deviation of the ball bearings is known to be 1.41 mm. Test, at the 2% significance level, whether the mean diameter of the ball bearings has changed, giving the critical region for the test. Assume that the diameters of the ball bearings are Normally distributed.

?

A certain company sells fruit juice in cartons. The amount of juice in a carton has a normal distribution with a standard deviation of 3ml.
The company claims that the mean amount of juice per carton, μ, is 60ml. A trading inspector has received complaints that the company is overstating the mean amount of juice per carton and wishes to investigate this complaint. The trading inspector takes a random sample of 16 cartons and finds that the mean amount of juice per carton is 59.1ml.

Using a 5% level of significance, and stating your hypotheses clearly, test whether or not there is evidence to uphold this complaint.

H_0 ?
H_1 ?
Let X be ? ?
Assuming H_0 ? ?
$N\left(60, 0.75^2\right)$
?

As always, state the distribution X and its distribution under H_0.

Use X to work out the distribution of the sample means, $\overline{X}$.

Note: Don't confuse X and $\overline{X}$. The X is the distribution over amounts of drink in each individual carton. $\overline{X}$ is the distribution over **sample means**, i.e. the possible sample means we see as we take samples of 16 cartons. X might not be normally distributed, but $\overline{X}$ will be.

Finding the critical region 80

[Textbook] A machine products bolts of diameter D where D has a normal distribution with mean 0.580 cm and standard deviation 0.015 cm. The machine is serviced and after the service a random sample of 50 bolts from the next production run is taken to see if the mean diameter of the bolts has changed from 0.580 cm. The distribution of the diameters of bolts after the service is still normal with a standard deviation of 0.015 cm.

(a) Find, at the 1% level, the critical region for this test, stating your hypotheses clearly.
The mean diameter of the sample of 50 bolts is calculated to be 0.587 cm.
(b) Comment on this observation in light of the critical region.

a ? $80, H_1 : \mu \neq 0.580$
Assuming H_0 ? ?
$\overline{D}$? $80, \left(\frac{0.015}{\sqrt{50}}\right)^2\right)$
?
$z = \pm 2.5758$
?
0.5745
$.5854$
$3sf)$
fficient evidence at
b ?

This is a two-tailed test so want the z value corresponding to the top 0.5% and bottom 0.5% respectively. Use table below or your calculator.

Use the usual formula $Z = \frac{X-\mu}{\sigma}$, except this time we use the standard deviation of $\overline{X}$: $\sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Edexcel S3 June 2011 Q7a

Roastie's Coffee is sold in packets with a stated weight of 250 g. A supermarket manager claims that the mean weight of the packets is less than the stated weight. She weighs a random sample of 90 packets from their stock and finds that their weights have a mean of 248 g and a standard deviation of 5.4 g.

(*a*) Using a 5% level of significance, test whether or not the manager's claim is justified. State your hypotheses clearly.

(5)

(a) ?

6 A machine produces metal bolts of diameter D mm, where D is normally distributed with standard deviation 0.1 mm. Bolts with diameter either less than 5.1 mm or greater than 5.6 mm cannot be sold.

Given that 5% of bolts have a diameter in excess of 5.62 mm,

a find the probability that a randomly chosen bolt can be sold. **(5 marks)**

Twelve bolts are chosen.

b Find the probability that fewer than three cannot be sold. **(2 marks)**

A second machine produces bolts of diameter Y mm, where Y is normally distributed with standard deviation 0.08 mm.

A random sample of 20 bolts produced by this machine is taken and the sample mean of the diameters is found to be 5.52 mm.

c Stating your hypotheses clearly, and using a 2.5% level of significance, test whether the mean diameter of all the bolts produced by the machine is less than 5.7 mm. **(4 marks)**

a?

b?

c?

7 The mass of European water voles, M grams, is normally distributed with standard deviation 12 grams.

Given that 2.5% of water voles have a mass greater than 160 grams,

a find the mean mass of a European water vole. **(3 marks)**

Eight water voles are chosen at random.

b Find the probability that at least 4 have a mass greater than 150 grams. **(3 marks)**

European water rats have mass, N grams, which is normally distributed with standard deviation 85 grams.

A random sample of 15 water rats is taken and the sample mean mass is found to be 875 grams.

c Stating your hypotheses clearly, and using a 10% level of significance, test whether the mean mass of all water rats is different from 860 grams. **(4 marks)**

a?

b?

c?

8 Daily mean windspeed is modelled as being normally distributed with a standard deviation of 3.1 knots.

A random sample of 25 recorded daily mean windspeeds is taken at Heathrow in 2015.

Given that the mean of the sample is 12.2 knots, test at the 2.5% level of significance whether the mean of the daily mean windspeeds is greater than 9.5 knots.

State your hypotheses clearly. **(4 marks)**

?

This is not in the textbook. But given the recent Chapter 2 on Conditional Probabilities and the fact that the type of question below occurred frequently in S1 papers, it seems worthwhile to cover!

Edexcel S1 May 2014(R) Q4

The time, in minutes, taken to fly from London to Malaga has a normal distribution with mean 150 minutes and standard deviation 10 minutes.

The time, X minutes, taken to fly from London to another city has a normal distribution with mean μ minutes.

Given that $P(X < \mu - 15) = 0.35$

(*c*) find $P(X > \mu + 15 \mid X > \mu - 15)$. **(3)**

?

Edexcel S1 Jan 2013 Q4a,c

The length of time, L hours, that a phone will work before it needs charging is normally distributed with a mean of 100 hours and a standard deviation of 15 hours.

(*a*) Find $P(L > 127)$.

(3)

Alice is about to go on a 6 hour journey. Given that it is 127 hours since Alice last charged her phone,

(*c*) find the probability that her phone will not need charging before her journey is completed.

(4)

(a) ?

(c) ?

IQ is distributed as $X \sim N\left(100, 15^2\right)$.
A person is considered to be 'pretty smart' if their IQ is at least 130.
Determine the median IQ of 'pretty smart' people.

?